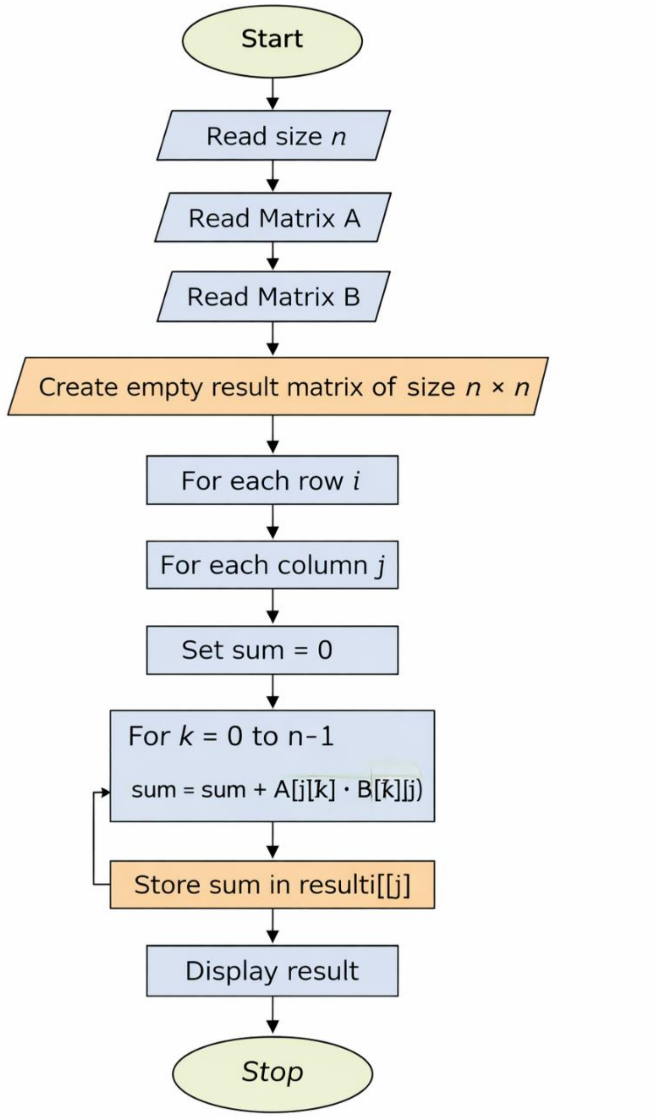
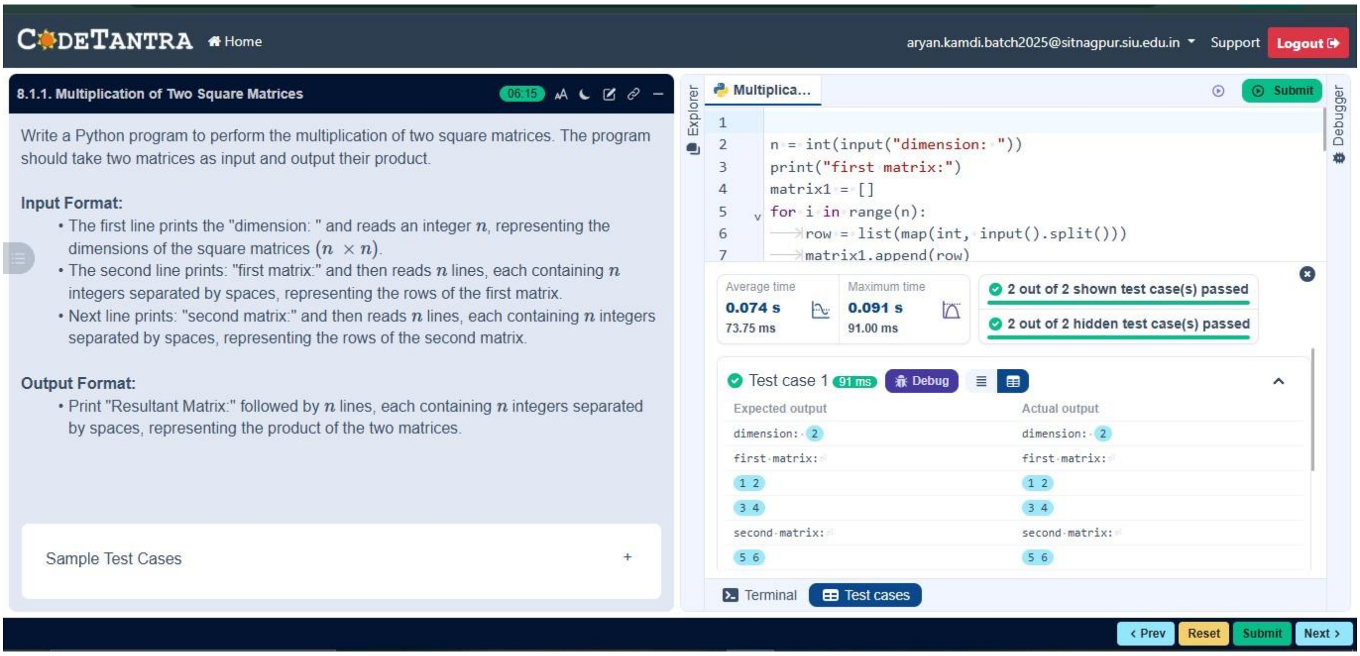
8.1.1 MULTIPLICATION OF 2 SQ. MATRICES

ALGORITHM

1. Start
2. Read size n
3. Read matrix A
4. Read matrix B
5. Create empty matrix result of size n × n
6. For each row i
7. For each column j
8. Set sum = 0
9. For k = 0 to n-1
10. sum = sum + A[i][k] × B[k][j]
11. Store sum in result[i][j]
12. Display result
13. Stop



Code: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8.1.1. Multiplication of Two Square Matrices
06:15
Write a Python program to perform the multiplication of two square matrices. The program should take two matrices as input and output their product.
Input Format:
• The first line prints the "dimension: " and reads an integer n, representing the dimensions of the square matrices (n × n).
• The second line prints: "first matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the first matrix.
• Next line prints: "second matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the second matrix.
Output Format:
• Print "Resultant Matrix:" followed by n lines, each containing n integers separated by spaces, representing the product of the two matrices.
Sample Test Cases
Explorer
Multiplica...
Submit
Debugger
n = int(input("dimension: "))
print("first matrix:")
matrix1 = []
for i in range(n):
row = list(map(int, input().split()))
matrix1.append(row)
Average time
0.074 s
73.75 ms
Maximum time
0.091 s
91.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
91 ms
Debug
Expected output
Actual output
dimension: 2
dimension: 2
first matrix:
first matrix:
1 2
1 2
3 4
3 4
second matrix:
second matrix:
5 6
5 6
Terminal
Test cases
‹ Prev
Reset
Submit
Next ›